



Pruning Trees

Ways to prevent overfitting

Decision trees are easy to overfit

- Early Stopping
- Pruning
- Ensembling

Decision Tree Pruning

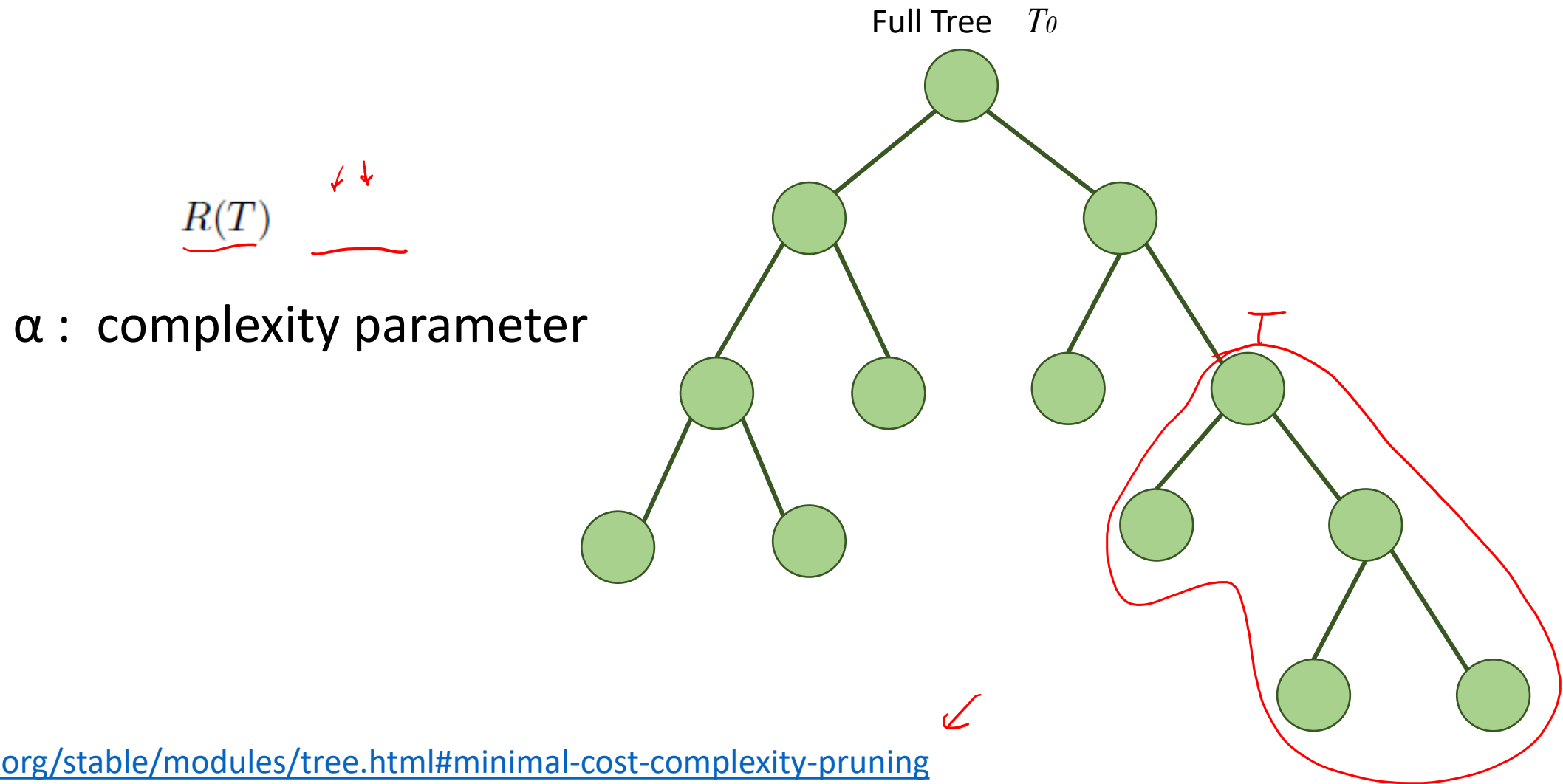
Issue: Sometimes a good split can happen later

Idea: Grow the tree fully then prune

How: Minimal Cost-Complexity Pruning

Pruning option is available since sklearn 0.22

Minimal Cost-Complexity Pruning



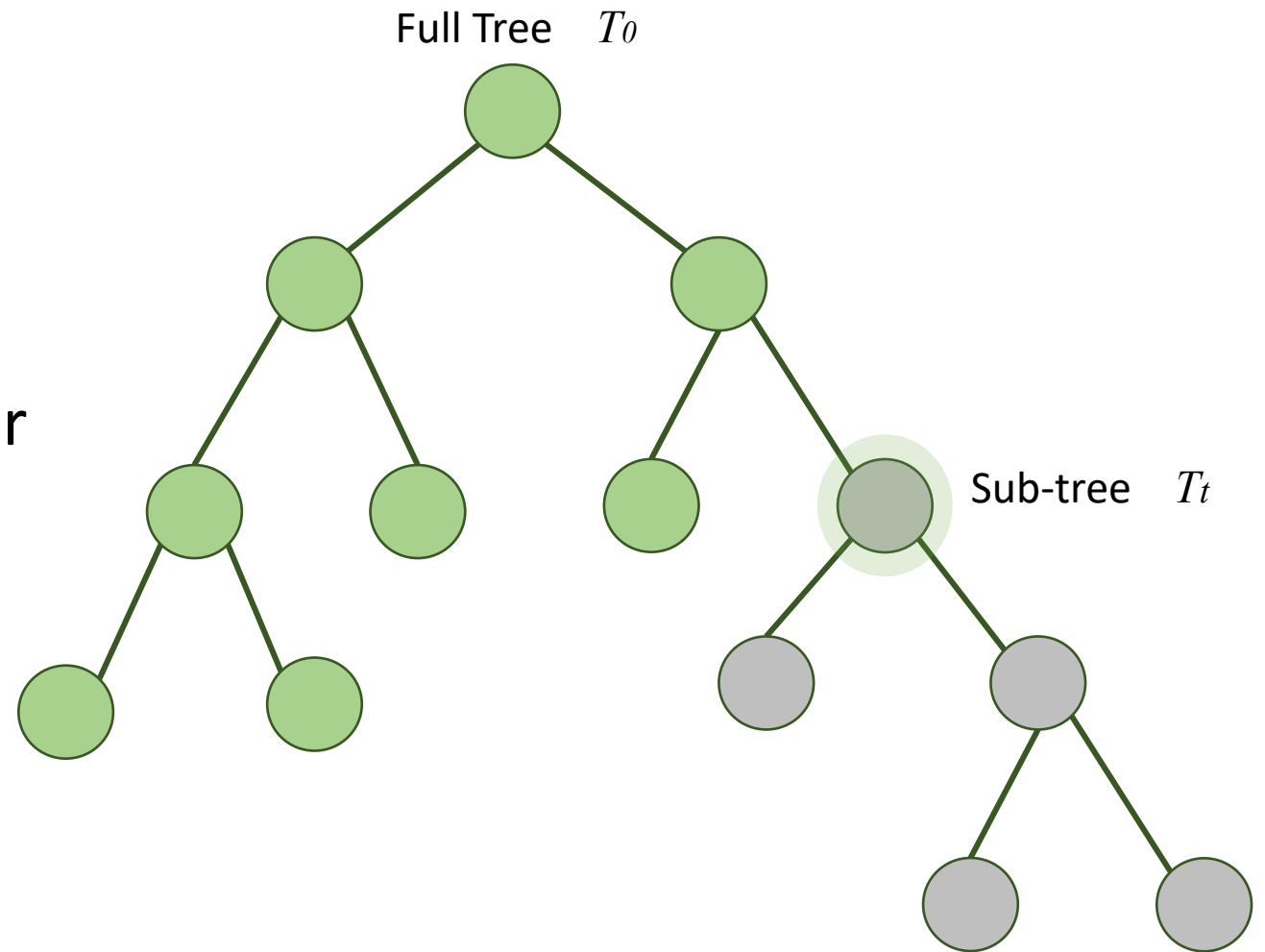
<https://scikit-learn.org/stable/modules/tree.html#minimal-cost-complexity-pruning>

https://scikit-learn.org/stable/auto_examples/tree/plot_cost_complexity_pruning.html

Minimal Cost-Complexity Pruning

$$R_{\alpha}(T) = R(T) + \alpha|T|$$

α : complexity parameter



Minimal Cost-Complexity Pruning

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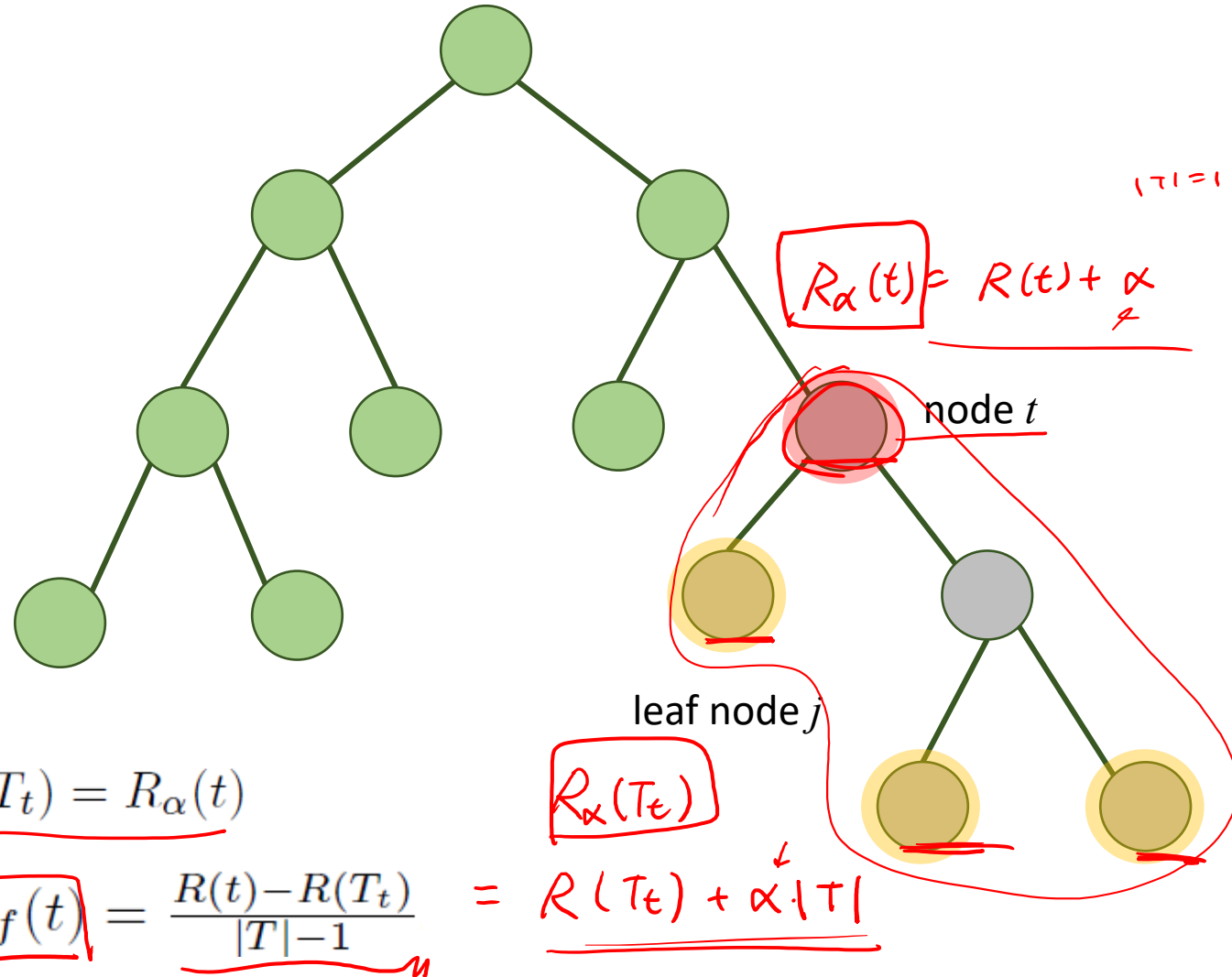
α : complexity parameter

$|T|$: number of leaf nodes of the subtree

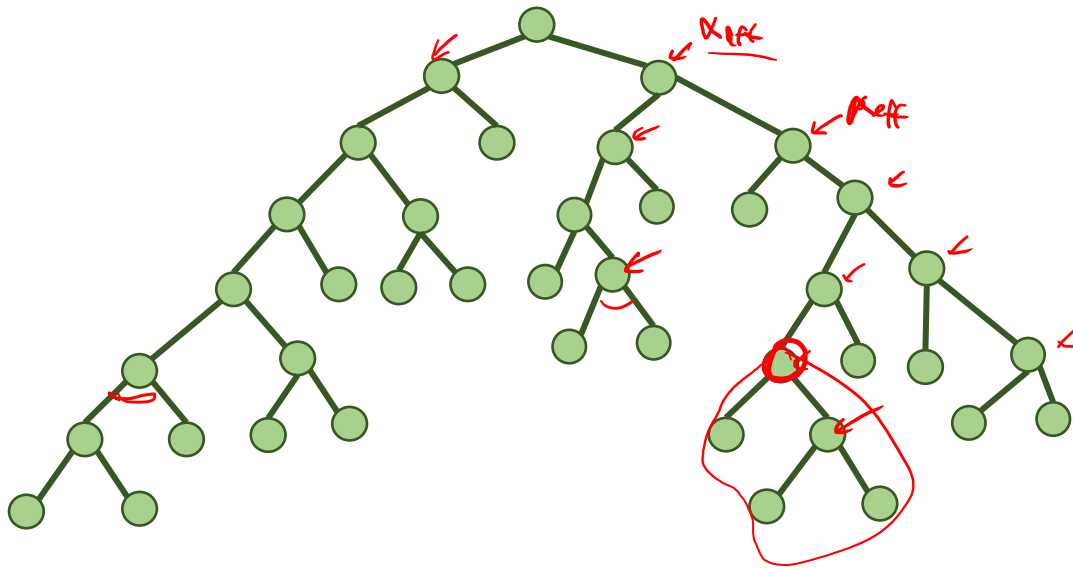
Impurity at the node t

$$R(T_t) < R(t)$$

Sum of the impurities
at the leaf nodes of the subtree T_t



Minimal Cost-Complexity Pruning



Iteratively removes the weakest link

When does it stop pruning?

Stop when $\min(\alpha_{eff}) > \underline{\alpha_{ccp}}$

α_{ccp} : cost complexity parameter,
"ccp_alpha"

Decision Tree Pros and Cons

Trees are easy to understand

Trees don't suffer collinearity

Trees are good for non-linear features

Trees handle categorical variables easily

Trees are weak-learner

Trees have high variance in general

Trees can overfit easily

Linear regression is a better choice if features are linear

Tree's performance can be greatly improved when **ensembled**